ARIMA Time Series Forecasting

Pakistan Rainfall & Flood Prediction

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COMP 4581 — Algorithms for Data Science

The Problem

2022 Pakistan Monsoon Floods

30+ million displaced

$30 billion in damages

1/3 of the country underwater

~10x normal monsoon rainfall

Can algorithms help predict this?

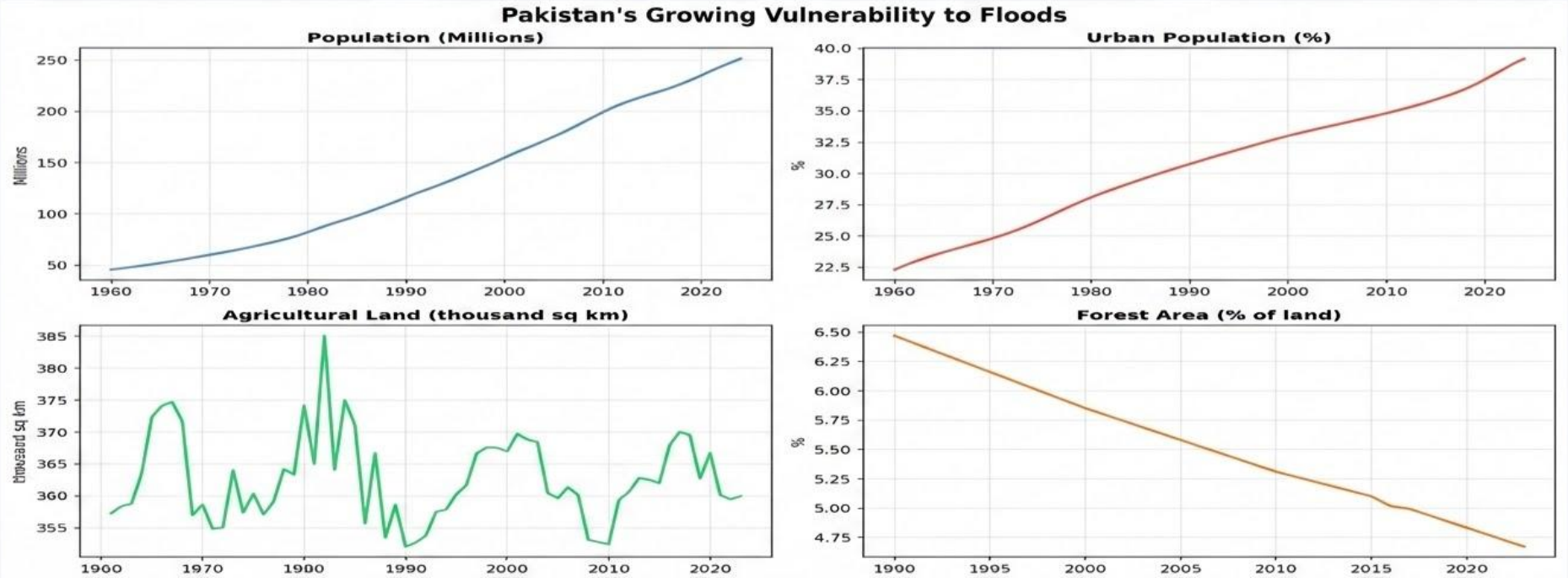
Growing Vulnerability

251M+
Population
5x since 1960

39%
Urban
2x since 1960

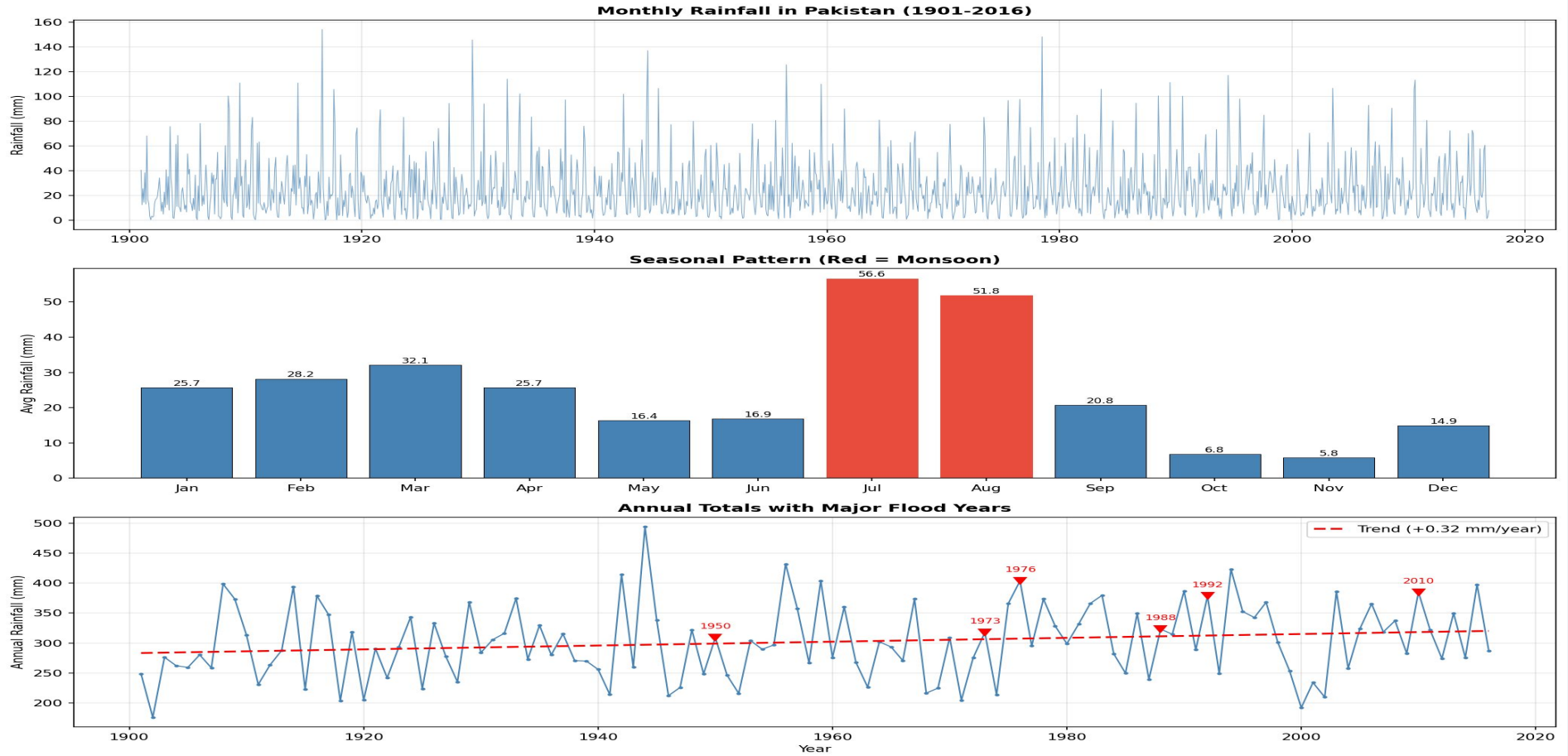
4.7%
Forest cover
down from 6.5%

$30B
2022 flood
damages



116 Years of Rainfall Data

1,392 monthly observations • clear monsoon spikes every July/August



How ARIMA Works

AR

Auto-Regressive

Uses past values to predict the next one

I

Integrated

Differencing for stationarity (d=0 for us)

MA

Moving Average

Uses past errors to self-correct

Complexity: O(n) per Kalman filter iteration

SARIMA

Adds seasonal parameters at lag 12

SARIMA(p,d,q)(P,D,Q,12)

Captures the monsoon cycle
that basic ARIMA misses

SARIMAX — same thing + external variable (ocean temps)

Before ARIMA: Is the Data Ready?

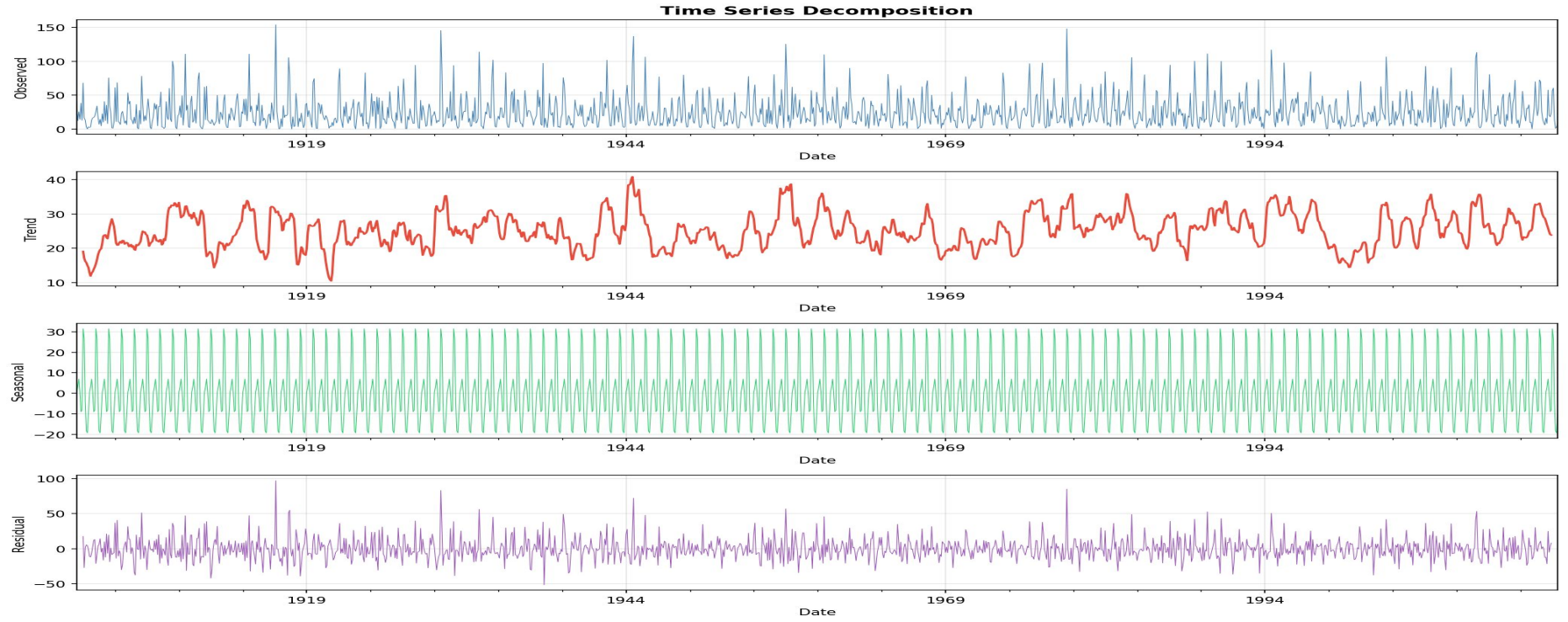
Dickey-Fuller: p ≈ 0.000 → Stationary (d=0)

ACF spike at lag 12

→ yearly seasonality confirmed

Clean seasonal pattern

→ SARIMA is a good fit



Results

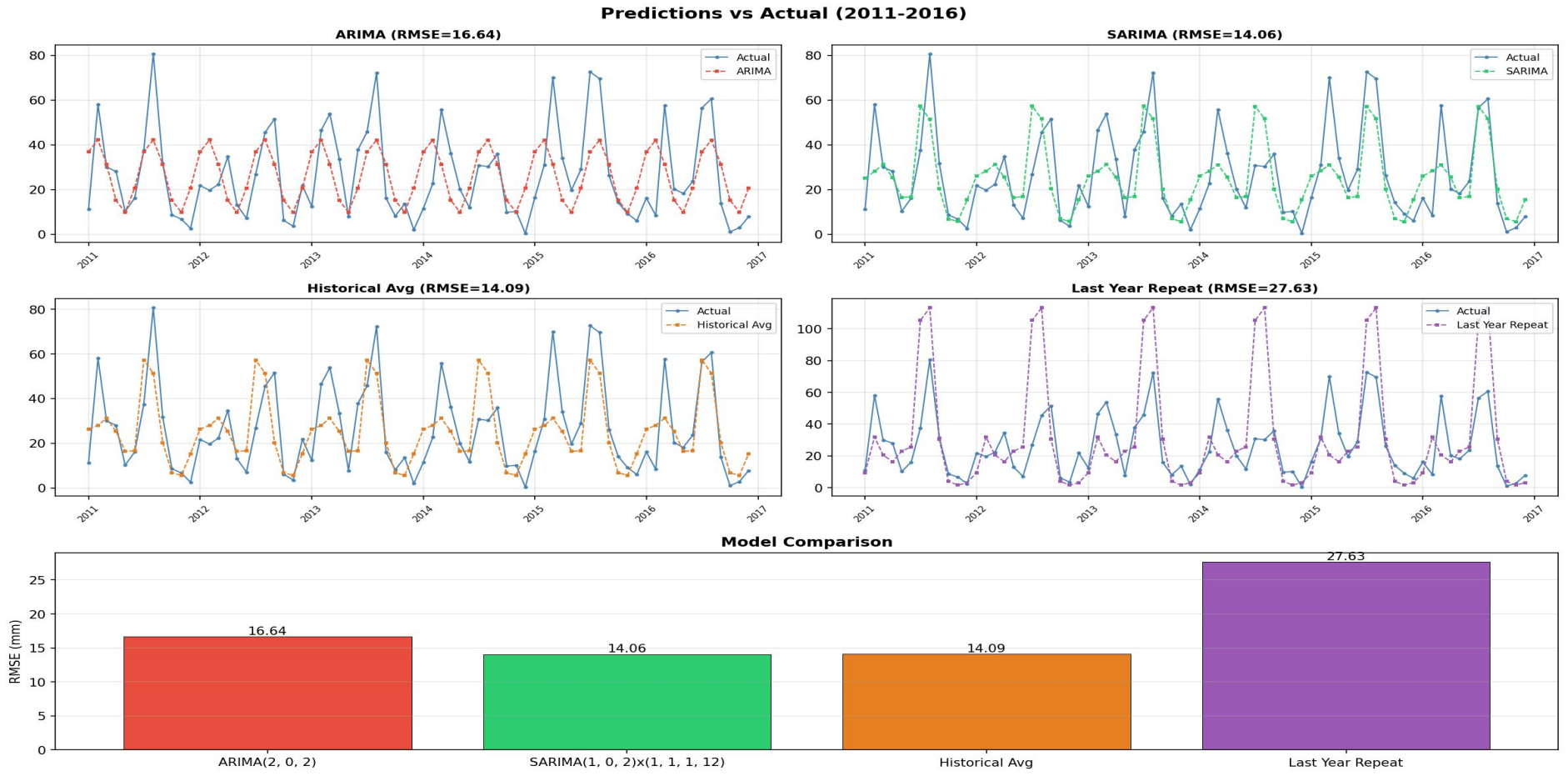
Trained on 1901-2010, tested on 2011-2016

Model	RMSE	MAE
ARIMA(2,0,2)	16.64	13.48
SARIMA(1,0,2)(1,1,1,12)	**14.06**	**10.98**
SARIMAX (+ ENSO)	13.70	10.62
Historical Average	14.09	11.01
Last Year Repeat	27.63	18.38

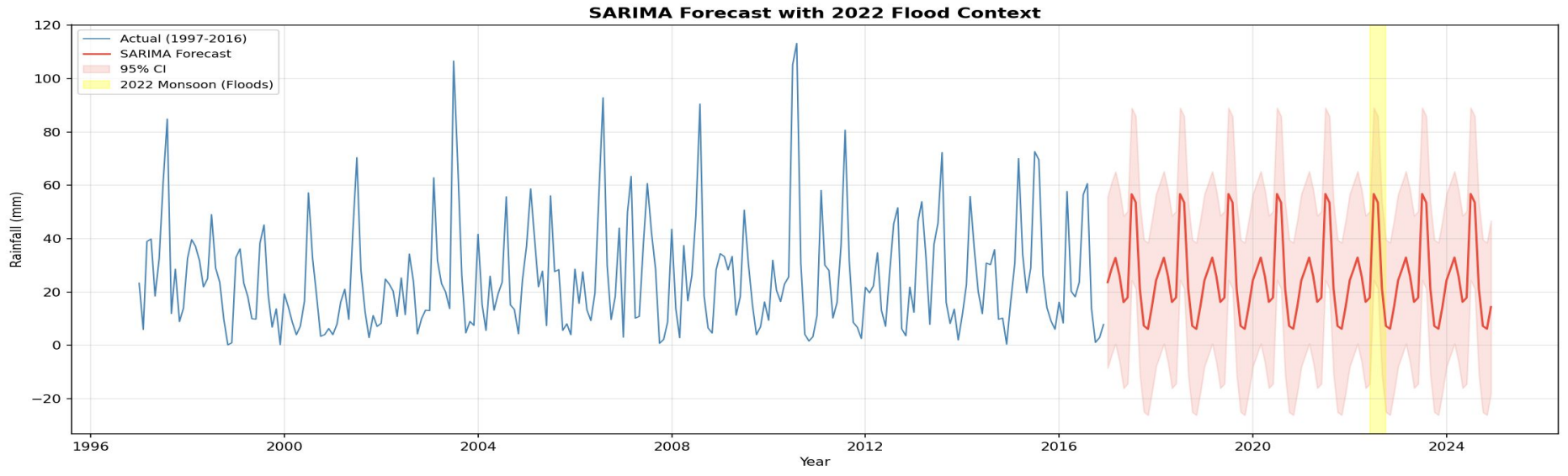
SARIMA wins — seasonal component is key. **ENSO barely helps** (0.1% improvement).

Last Year Repeat is worst — confirms the algorithm adds real value.

Predictions vs Actual (2011-2016)



Could This Have Predicted 2022?



Predicted: ~57 mm for July 2022

95% CI: 24–89 mm

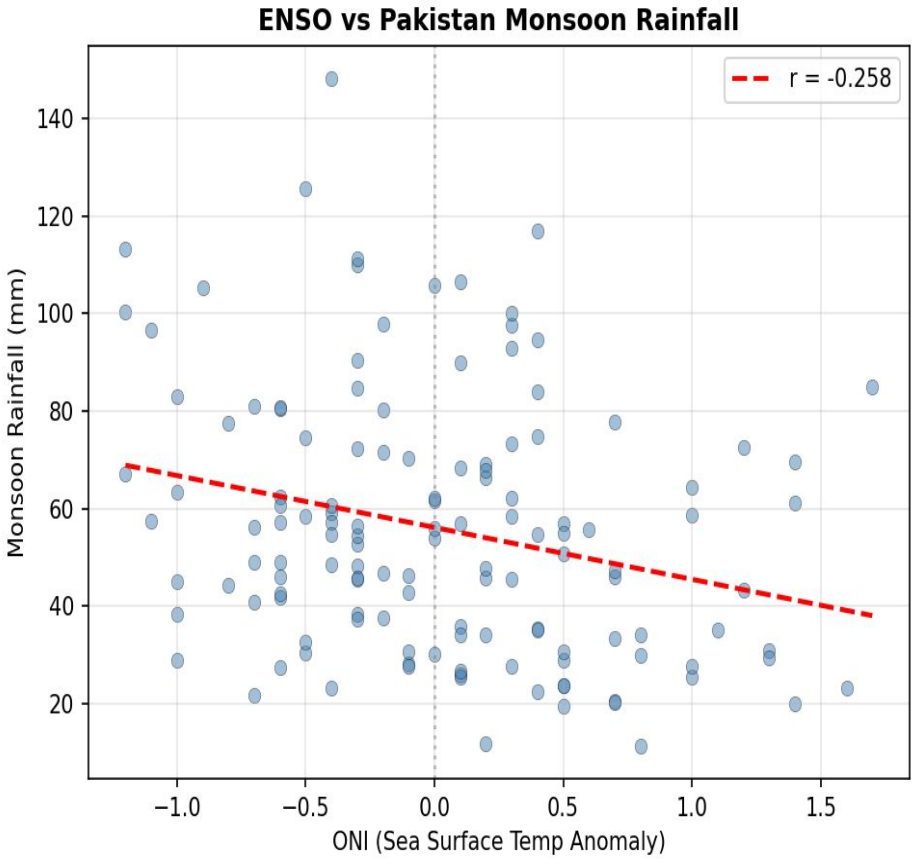
Actual: ~10x that amount

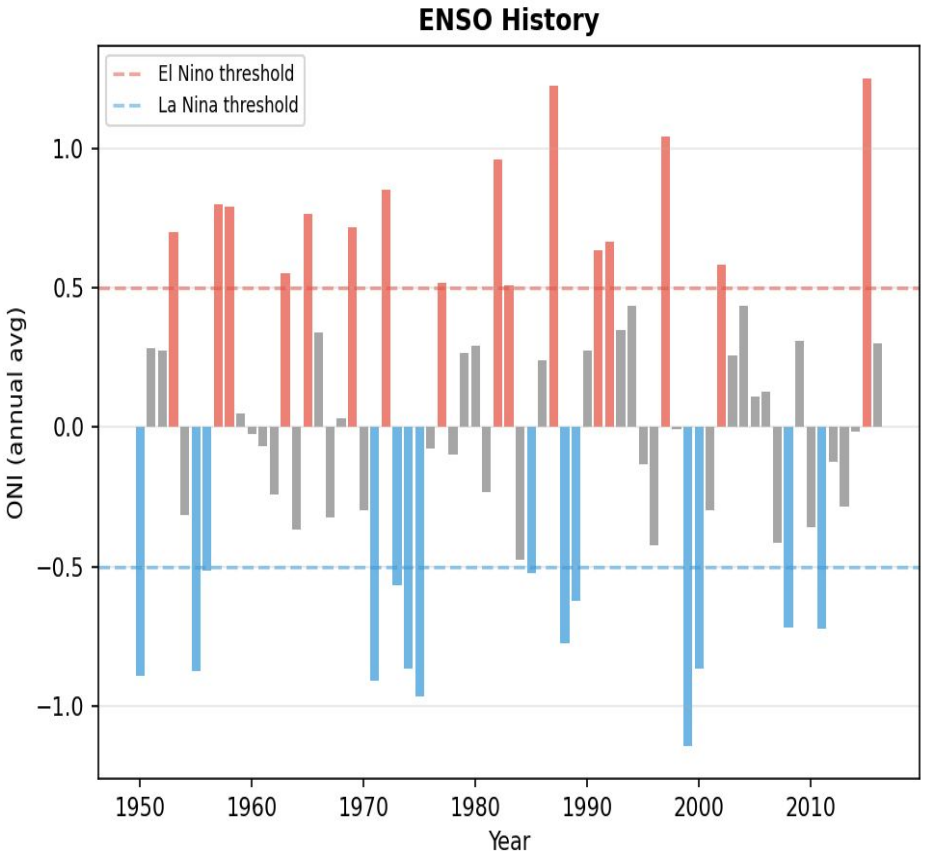
But that's the point.

ARIMA defines "normal." Readings far above the confidence interval ARE the alarm.

Testing Ocean Temperatures (ENSO)

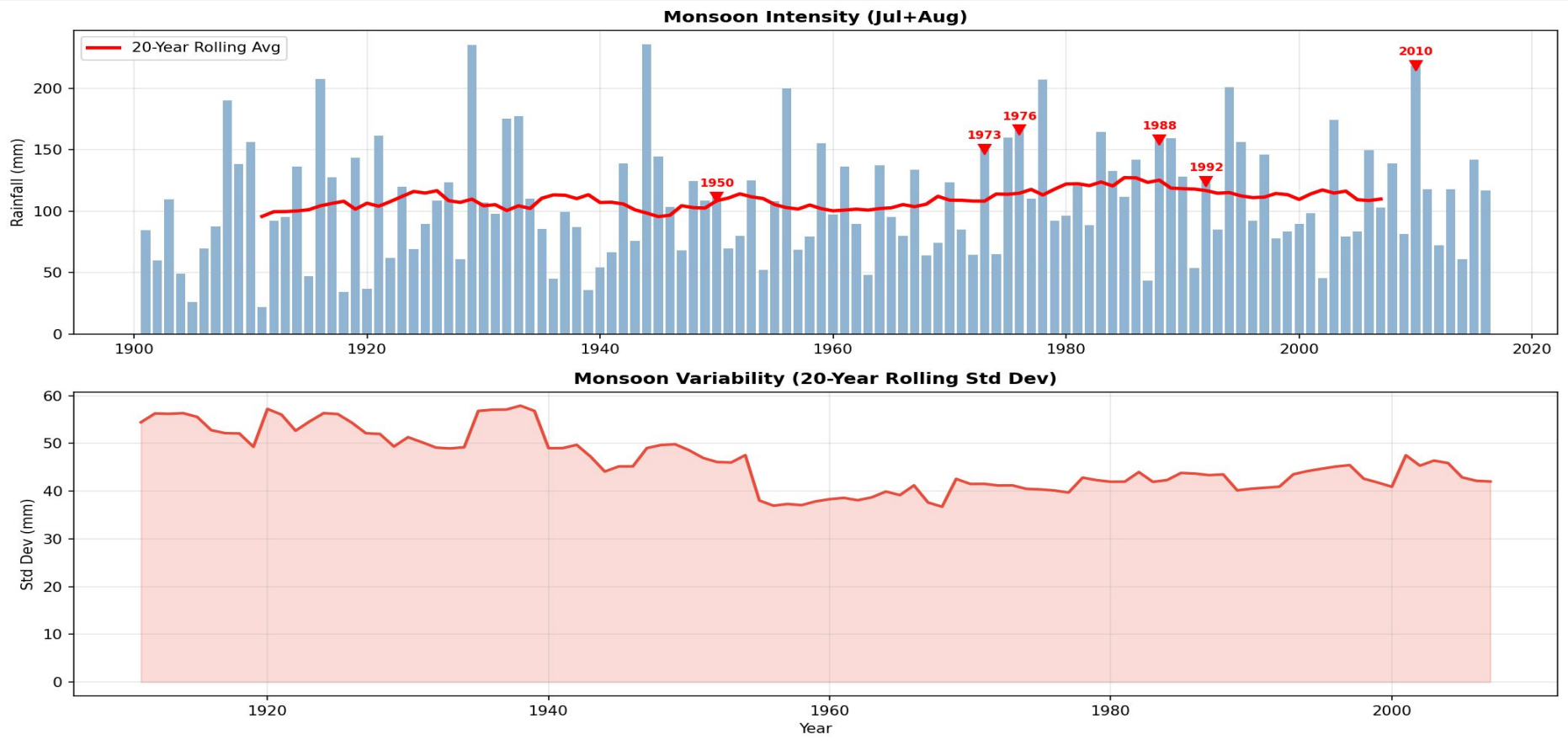
r = -0.258 ⟶ La Niña brings more rain (expected), but only 0.1% RMSE improvement





Monsoon Intensity: 116 Years

Variability is increasing — monsoon seasons are becoming less predictable



Limitations & Future Work

What it can't do:

Linear model — learns patterns, not physics

2022 was unprecedented in the record

ENSO helps at seasonal scale, not monthly

Other factors untested (glacial melt, IOD)

What's next:

ARIMA + LSTM ensemble models

Satellite imagery for real-time alerts

Seasonal forecasting instead of monthly

Regional breakdown (Sindh, Punjab, KPK)

Questions!

3 Datasets • 5 Models • 116 Years

Namoos Haider | COMP 4581 | Spring 2026